

# Photonics Heterogeneous Integration (PHI) of Thin-Film Lithium Niobate and Hydrogen-Free Silicon Nitride on a 200-mm Silicon Photonics Platform

L. Tobing<sup>1,2</sup>, B. Li<sup>1,2</sup>, B. Pan<sup>1</sup>, C. Lau<sup>1</sup>, A. W. K. Fong<sup>1</sup>, Y. Zhang<sup>1</sup>, Y. Cao<sup>1</sup>, J. Zhou<sup>2</sup>, S. N. Foo<sup>1,2</sup>, W. Q. Tan<sup>1,2</sup>, C. Lee<sup>1,2</sup>, J. O. Yoo<sup>1,2</sup>, F. Masahisa<sup>1,2</sup>, L. Xie<sup>2</sup>, N. Venkataraman<sup>1,2</sup>, S. C. Chong<sup>2</sup>, T. Ghosh<sup>1,2</sup>, M. Zhao<sup>3</sup>, D. Zhu<sup>1,3</sup>, K. W. Ang<sup>1</sup>, N. Singh<sup>1,2</sup>, L. W. Lim<sup>1,2,\*</sup>, X. Luo<sup>1,2,\*</sup>, Y.-C. Yeo<sup>1,\*</sup>

<sup>1</sup>National Semiconductor Translation and Innovation Centre (NSTIC), 4 Fusionopolis Way, Kinesis #09-11, Singapore 138635

<sup>2</sup>Institute of Microelectronics (IME), A\*STAR, 2 Fusionopolis Way, Innovis #08-02, Singapore 138634

<sup>3</sup>Department of Materials Science and Engineering, National University of Singapore, Singapore 117575

\* lim\_leh\_oon@a-star.edu.sg; luo\_xianshu@a-star.edu.sg; yeo\_yee\_chia@nstic.sg

**Abstract:** We demonstrate 200-mm wafer-scale photonic heterogeneous integration platform comprising TFLN modulator and hydrogen-free SiN using die-to-wafer bonding. High-efficiency light transmission between SiN and TFLN waveguides is demonstrated using MZI modulator with 2.9 V·cm modulation efficiency. © 2026 The Author(s)

## 1. Introduction

Silicon photonics technology has successfully demonstrated 200 G/lane data rate with recent product ramp. To achieve 400 G/lane or higher, it may be challenging to surmount intrinsic limitations of silicon modulators (e.g. balancing optical modulation speed, efficiency, and insertion loss). Therefore, heterogeneous integration of the best-in-class materials has become an essential pathway for further performance enhancements. Materials are selected for their superior performance in specific photonic functionality such as low-loss SiN waveguides, high-speed thin-film lithium niobate (TFLN) modulators, and III-V optoelectronic devices. To this end, die-to-wafer (D2W) bonding, which has been successfully commercialized for optical transceivers with hybrid laser integration [1], is a promising approach for its compelling advantages in the scalability to 200 mm and 300 mm platform, versatility of integrating diverse materials on silicon photonic wafer, CMOS-compatible manufacturability, and reduced material cost.

In this report, we show the first demonstration of photonic heterogeneous integration (PHI) of TFLN modulators on low-loss hydrogen-free silicon nitride passive waveguides through D2W bonding on a 200 mm silicon photonics platform. We successfully demonstrate low-loss optical transmission between SiN and TFLN with 0.4 dB/transition and propagation losses of 0.4 dB/cm and 1.2 dB/cm for SiN and TFLN respectively. TFLN-on-SiN MZI modulator shows 2.9 V·cm modulation efficiency. While TFLN-on-Si modulators have been demonstrated [2-3], this work represents the first report of 200-mm wafer-scale TFLN-on-SiN integration using D2W bonding, with subsequent post-bond TFLN die patterning for modulator device fabrication. The demonstrated wafer-scale PHI, enabled by commercially mature high-yield high-throughput D2W bonding, integrates novel materials on silicon photonics wafers for enhanced functionalities and device performances, paving the way towards fully integrated 400 G/lane optical transceiver with scalability and manufacturability.

## 2. Photonic Heterogeneous Integration (PHI) via Die-to-Wafer Bonding

Fig. 1 presents the proposed PHI process flow, which comprises 3 process modules: (1) passive waveguide patterning and planarization, (2) D2W bonding of the TFLN dies and substrate removal, and (3) post-bond TFLN patterning and metallization for modulators. These process modules are optimized to achieve cost-effectiveness and manufacturability of the PHI process flow.

### 2.1. Passive waveguide fabrication

SiN is chosen as the waveguide material for its low optical loss, low thermal sensitivity, wide fabrication tolerance, and integration flexibility. To effectively eliminate optical absorption related to Si-H/N-H bonds without high temperature annealing, we developed a novel hydrogen-free low-thermal-budget near-stoichiometric SiN process using deuterated silane (SiD<sub>4</sub>) as the precursor. Using machine-learning assisted recipe optimization, we etched SiN waveguide with line-edge-roughness (LER) of ~1 nm [Fig. 2(a)]. Fig. 2(b)-(c) show the propagation loss characteristics of SiN:D waveguide in C- and O-bands, with optical loss of around 0.41 dB/cm at 1550 nm and 0.21 dB/cm at 1310 nm without thermal annealing.

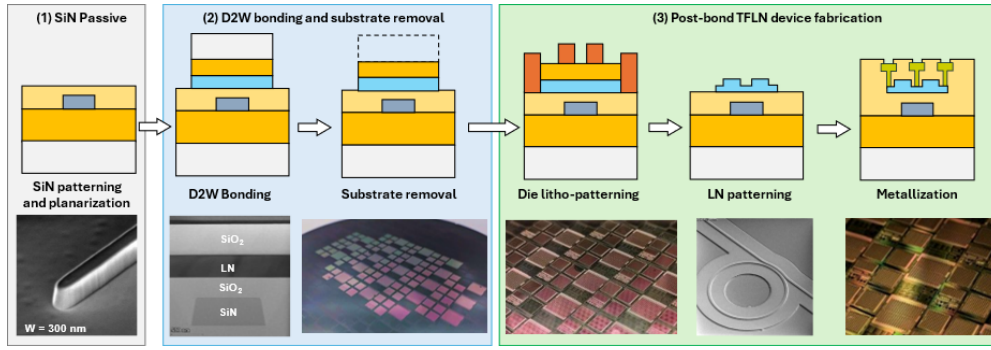


Fig. 1. PHI process flow including SiN passive, D2W bonding, and TFLN modulator fabrication.

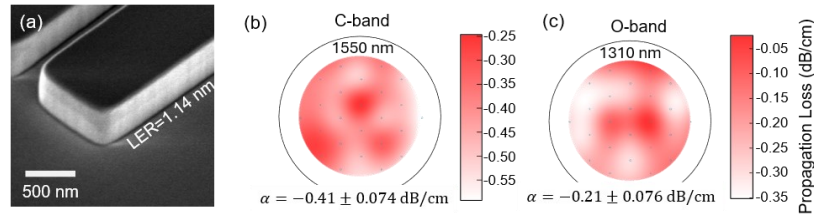


Fig. 2. (a) Representative SEM micrograph for the fabricated SiN:D waveguide. (b)-(c) Wafer-scale propagation loss at 1550 nm and 1310 nm, without thermal annealing.

## 2.2. Wafer-scale D2W bonding

TFLN-on-insulator wafers with a 725  $\mu\text{m}$ -thick silicon substrate are diced into 10 mm  $\times$  10 mm and 6 mm  $\times$  6 mm dies. Various optimizations of the D2W bonding process are done to ensure high yield for subsequent TFLN modulator fabrication: wafer dicing with minimal chipping, die thickness and post-bond thermal treatment optimization. Silicon substrate is then removed. The optical micrograph after substrate removal is shown in Fig. 1, where 350-nm thick TFLN with 500 nm-thick thermal oxide film have been successfully bonded. High resolution TEM of the TFLN-SiO<sub>2</sub> interface confirms high quality and uniform bonding interface. Confocal Scanning Acoustic Microscopy (C-SAM) confirms relatively good bonding yield.

## 2.3. Post-bonding TFLN device fabrication

PHI-LN devices including passive TFLN waveguides and modulators are then fabricated using our in-house 200 mm BEOL fabrication facilities. Integration process issues associated with TFLN are addressed, including TFLN etch and its non-volatile byproduct removal, lithography with topography induced by the bonded TFLN dies, and handling of potential lithium contamination. With the optimized wafer handling protocol and integration process, an integration flow for TFLN modulator fabrication is established. A single-layer metallization scheme comprising Aluminum for via fill and Ti/TiN as adhesion/barrier layer is used. Despite the 1  $\mu\text{m}$  step height from substrate removed dies, the lithography process remained well-controlled, with good alignment overlay of the TFLN waveguide to the underlying SiN waveguide (a key advantage of the D2W-based PHI compared to micro-transfer printing with  $> 10\times$  worse alignment accuracy [4]). D2W bonding also enables spot-size-converter (SSC) designs with high vertical coupling efficiency. Fig. 3 presents SEM images of fabricated devices, including LN waveguides, SiN/LN SSC, MZI and ring modulators, and the TEM of the MZI modulator. For MZI modulator, all passive functionalities such as light input/output, MMI splitters, waveguide bends are performed in SiN, while only straight TFLN waveguide is used for the light modulation.

## 3. Device characterization

Wafer-scale characterization of passive and active devices is performed in the C / L band. There is negligible impact on the SiN loss after bonding and TFLN processing, as evident from the comparison of pre- and post-PHI processing [Fig. 4(a)]. Wafer-scale TFLN waveguide loss, extracted from TFLN ring resonators, is plotted in Fig. 4(b), showing an average loss of  $2.4 \pm 0.89$  dB/cm, and a minimum of 1.2 dB/cm. These values are higher than both our 200 mm TFLN-on-insulator baseline and state-of-the-art TFLN-on-insulator waveguides ( $< 0.5$  dB/cm) [5], suggesting the need for further process optimization in the PHI process. Fig. 4(c) presents transmission spectrum of 70  $\mu\text{m}$ -radius LN ring coupled from SiN passive waveguide via an SSC structure, with the highest loaded Q-factor of  $\sim 5.5 \times 10^4$ . The SSC insertion loss between TFLN and SiN is  $\sim 0.4$

dB/transition. Fig. 4(d) shows the spectral response of TFLN MZI with unbalanced arms in SiN layer, exhibiting up to  $\sim 40$  dB extinction ratio, where the light is coupled via SiN grating couplers. Fig. 4(e) shows the DC response of the TFLN MZI modulator with 7 mm arm length and 4  $\mu$ m metal gap with  $V_{\pi}L$  of 2.9 V $\cdot$ cm. Fig. 4(f) shows the EO response of the same MZI modulator, exhibiting a 3-dB bandwidth of  $\sim 20$  GHz.

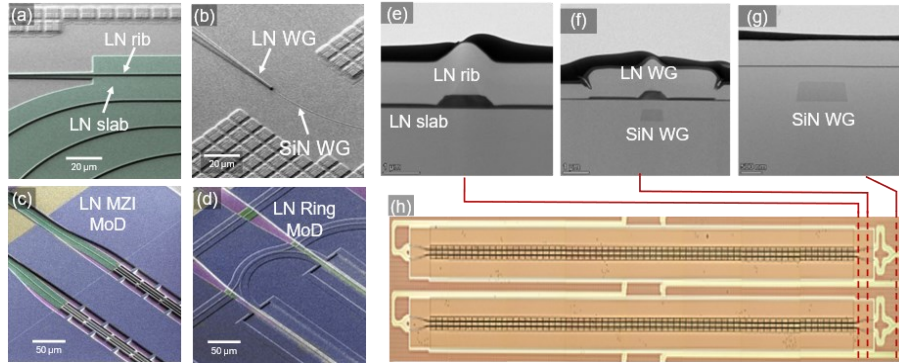


Fig. 3. SEMs of the fabricated devices: (a) Rib/Slab TFLN waveguides, (b) TFLN-SiN spot-size converter (SSC), (c)-(d) MZI and microring modulators after metallization, (e)-(g) TEM images of the TFLN-SiN SSC transition in TFLN WG, SiN-TFLN waveguide, and SiN waveguide areas, (h) Optical micrograph of a PHI-LN MZI modulator.

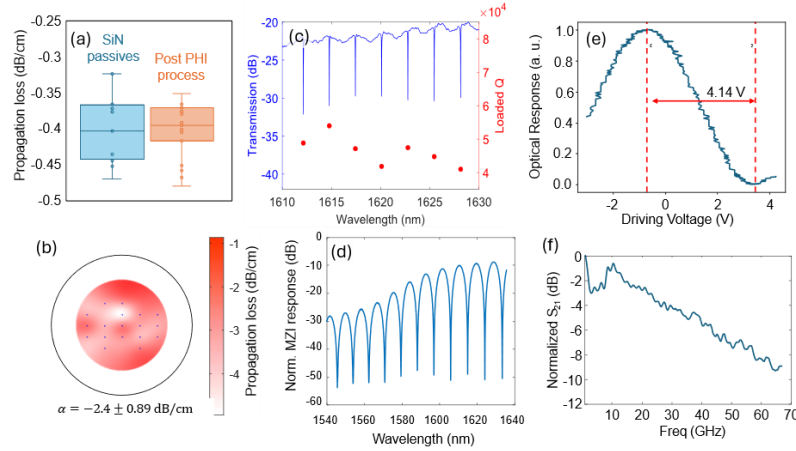


Fig. 4. Characterization of PHI-LN devices: (a) Comparison of the loss at 1550 nm before and after PHI process, (b) extracted wafer-scale TFLN waveguide loss, (c) TFLN microring transmission spectrum and the extracted loaded Q-factor, (d) TFLN MZI spectrum, (e) DC response of a TFLN MZI modulator of 7 mm length and 4  $\mu$ m metal gap with  $V_{\pi}L = 2.9$  V $\cdot$ cm, (f) EO response of TFLN modulator of 7 mm length and 4  $\mu$ m metal gap with 3-dB bandwidth of 20 GHz.

#### 4. Conclusion

We report the first demonstration of photonics heterogeneous integration of TFLN modulators with hydrogen-free SiN waveguides through die-to-wafer bonding on 200-mm silicon photonics platform. Relatively high bonding yield is achieved with efficient optical light transmission between SiN and TFLN waveguides. The SiN waveguide maintains low loss ( $\sim 0.4$  dB/cm in the C-band) after post-bond processing, while TFLN modulators exhibits minimum waveguide loss of 1.2 dB/cm and DC modulation efficiency of 2.9 V $\cdot$ cm. Preliminary RF characterization on the TFLN modulators shows a 3-dB bandwidth of 20 GHz.

#### Acknowledgments:

We acknowledge the funding support from National Semiconductor Translation and Innovation Centre.

#### 5. References

- [1] F. Vallini, et al., Proc. SPIE 13371, 1337106, Mar. 2025.
- [2] F. Valdez, et al., Sci Rep, 12, 18611, 2022.
- [3] M. He, et al., Nature photonics 13, pp. 359-364, 2019.
- [4] G. Roelken, et al., APL Photonics, 9, 010901, 2024.
- [5] D. Zhu, et al, Advances in Optics and Photonics 13, 242-352, 2021.